



## MY-CAM004M Product Manual



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**Shenzhen Mier Electronics Co., Ltd.**

# Version History

| Version | Authors and participants' dates |  |          | Remark              |
|---------|---------------------------------|--|----------|---------------------|
| V1.0    | Sender                          |  | 20240206 | Officially released |

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## 1. Product Introduction

The MY-CAM004M is a video conversion module launched by Shenzhen MYIR Electronics Co., Ltd., used for 4-channel AHD to MIPI conversion.

This module allows users to evaluate the interface functions and performance of MIRV development board products. It uses the N4 chip paired with basic peripheral circuitry.

In project development based on the MIRV development board for camera display, users can purchase this module and pair it with 1-4 high-definition cameras for display and monitoring.

Embedded development can reduce the development cycle and cost.

The N4 chip supports single-channel and virtual-channel MIPI-CSI2, 4-lane; it supports MIPI-CSI2 1/2/4 data channel configurations. It supports 2...

Data channels up to 720p@50/60, 1080p@25/30 2CH; supports 4 data channels up to 720p@50/60, 1080p@25/30

4CH. This module comes with a dedicated camera, model: QJD-SONY 225, AND: 200W, effective pixels: 1920 (H) ×

For details on 1080 (V), please consult sales.

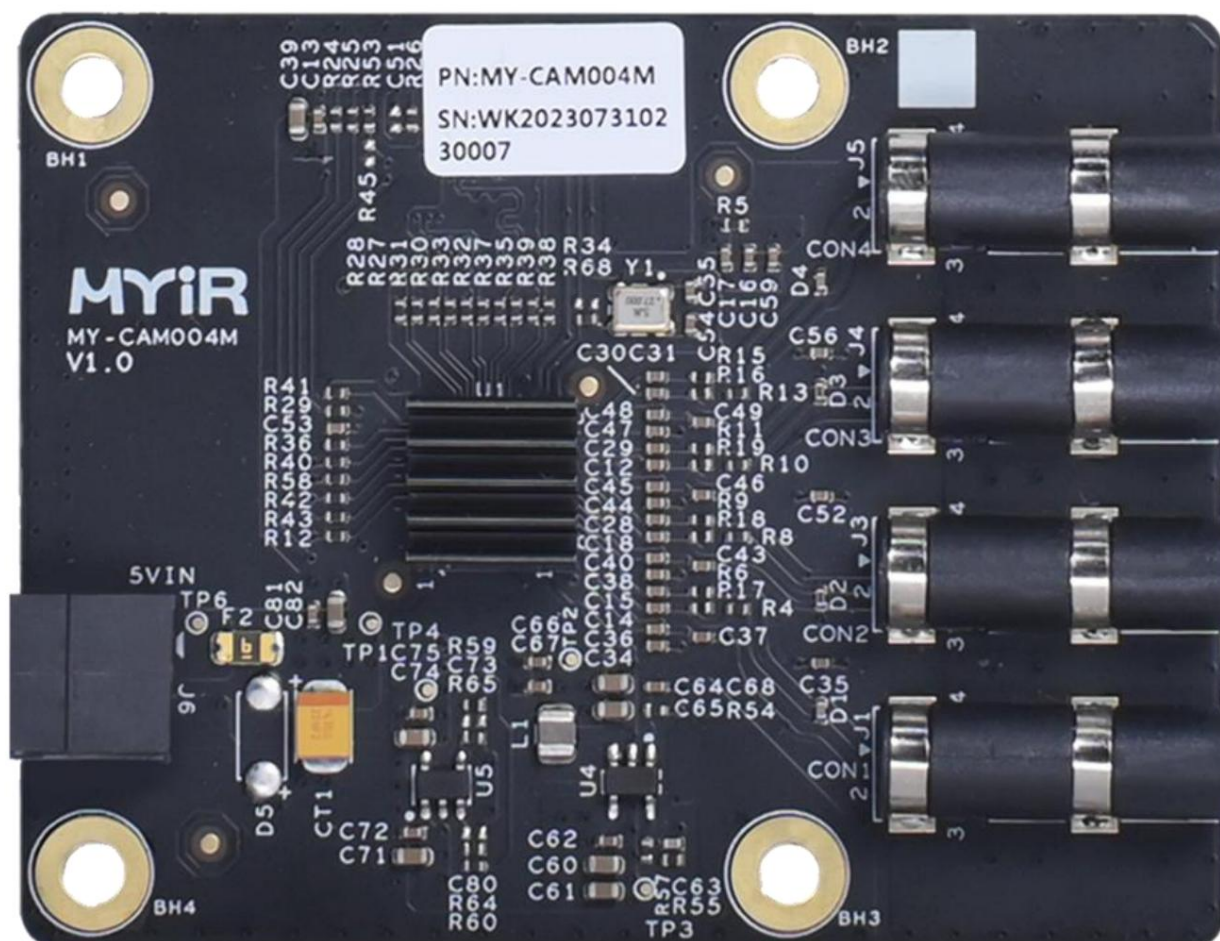


Figure 1-1 Front view of MY-CAM004M



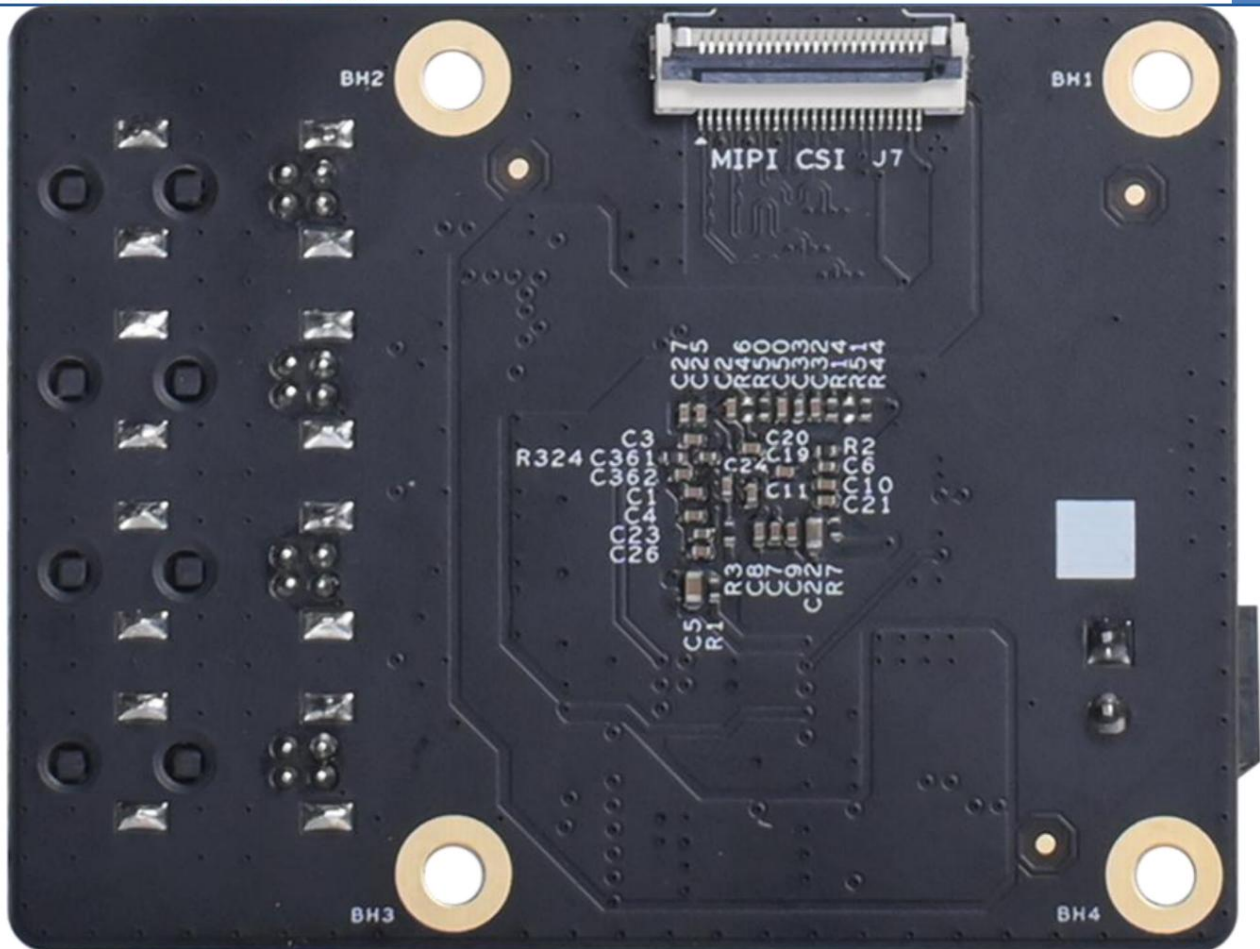


Figure 1-2 Back view of MY-CAM004M

### 1.1. Introduction to the N4 Chip

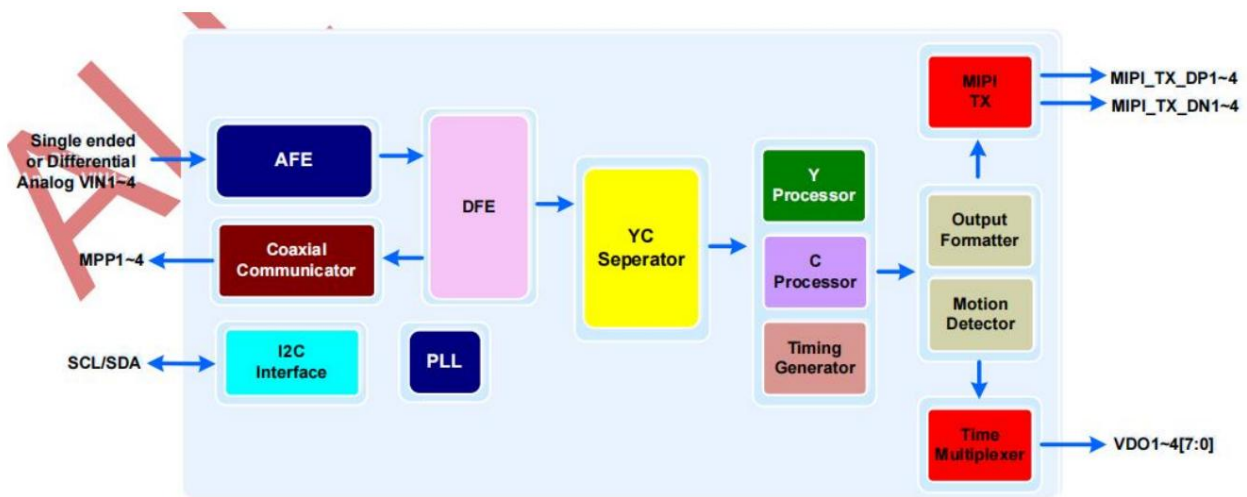
**N4** includes Automotive4-Channel RX with MIPI-CSI2/BT.656 Interface. It delivers high quality CVBS, 1M, 2M image. It accepts CVBS, COMET, 1M, 2M analog inputs from Camera and the other video signal sources. It accepts Single-ended/Differential analog SD/HD/FHD video signals, and then Processing Clamp/AGC(Auto Gain Control)/YC Separation and Converting BT.656/BT.1120/MIPI-CSI2 Format. MIPI CSI2 interface compliant with MIPI D-PHY v1.1.

**N4** is able to use same transmission cable with conventional one for CVBS/COMET(SD level), 1M/2M(HD/FHD level) and they provide the superior-image quality by minimizing the interference when separating Y and C.





N4support Coaxial Communication Protocolcommunicates between controller(DVR) and camera on the video signal through coaxial cable.



## 2. Interface Layout

The MY-CAM004M uses a double-sided component layout design. Key IC components and connectors are labeled below.

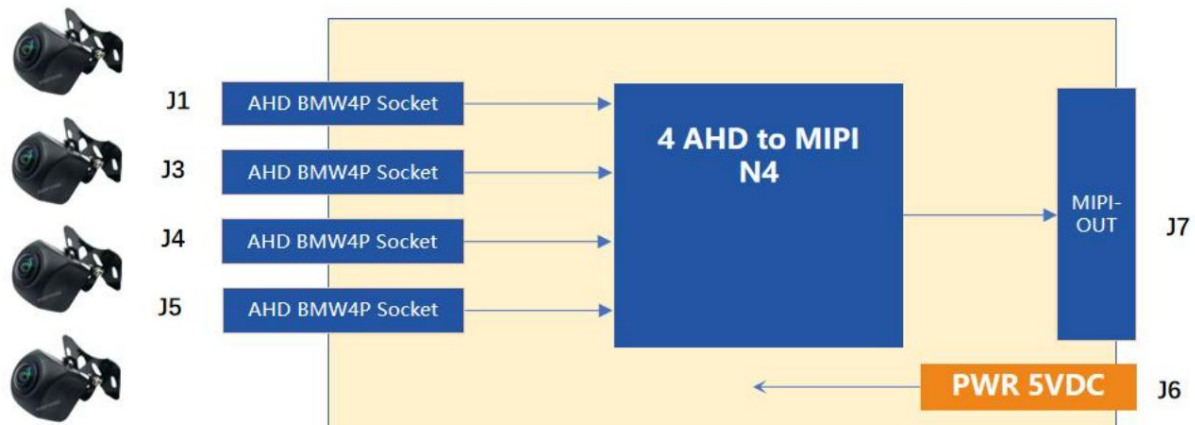


Figure 2-1 MY-CAM004M Annotation Diagram



3. Interface Description

3.1. CSI Interface

The MY-CAM004M's CSI interface uses a 24-pin FPC connector. This connector features a flip-top design, bottom contact, and a 0.5mm pin. Spacing. Reference part number: 1013A0024-1P1, brand: GDZ. Compatible heatsink for the chip: MY-CAM004M-RADIATOR-NA-V01ÿ

When connecting the FPC flexible flat cable to this module, ensure that the FPC cable pins are inserted downwards into the FPC connector.

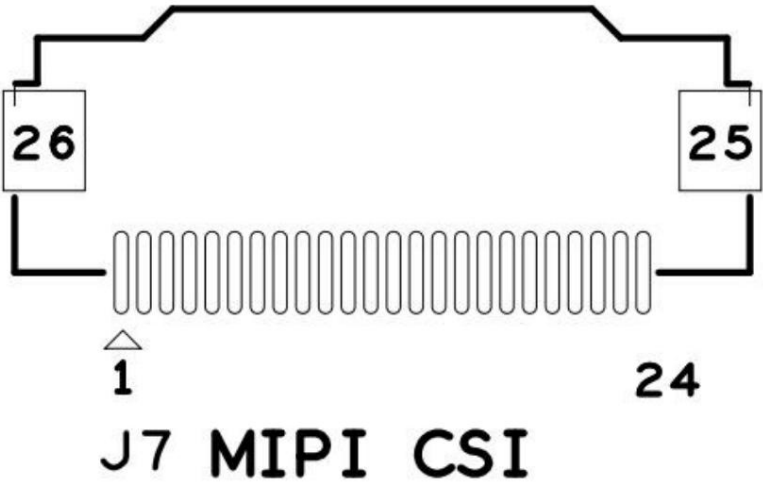


Figure 3-1 Pinout Diagram of MY-CAM004M

3.1.1. Pin Definitions

| Position | logo | Function                               | Signal          | illustrate |
|----------|------|----------------------------------------|-----------------|------------|
| J7       | 1    | CSI Differential Negative Signal       | MIPI_CSI1_D0_N  |            |
|          | 2    | CSI Differential Positive Signal       | MIPI_CSI1_D0_P  |            |
|          | 3    | GND                                    | GND             |            |
|          | 4    | CSI Differential Negative Signal       | MIPI_CSI1_D1_N  |            |
|          | 5    | CSI Differential Positive Signal       | MIPI_CSI1_D1_P  |            |
|          | 6    | GND                                    | GND             |            |
|          | 7    | CSI Clock Differential Negative Signal | MIPI_CSI1_CLK_N |            |
|          | 8    | CSI Clock Differential Positive Signal | MIPI_CSI1_CLK_P |            |
|          | 9    | GND                                    | GND             |            |
|          | 10   | CSI Differential Negative Signal       | MIPI_CSI1_D2_N  |            |
|          | 11   | CSI Differential Positive Signal 6     | MIPI_CSI1_D2_P  |            |



|  |    |                                         |                                                |           |
|--|----|-----------------------------------------|------------------------------------------------|-----------|
|  | 12 | GND                                     | GND                                            |           |
|  | 13 | CSI Differential Negative Signal        | MIPI_CSI1_D3_N                                 |           |
|  | 14 | CSI Differential Positive Signal        | MIPI_CSI1_D3_P                                 |           |
|  | 15 | GND                                     | GND                                            |           |
|  | 16 | CSI clock signal                        | MIPI_CSI1_MCLK                                 | 3V3 level |
|  | 17 | GND                                     | GND                                            |           |
|  | 18 | CSI Enable Signal                       | MIPI_CSI1_PWRDN is pulled down 4.7K to ground. |           |
|  | 19 | CSI Reset Signal                        | MIPI_CSI1_RST_N                                | 3V3 level |
|  | 20 | CSI 12C clock signal                    | MIPI_CSI1_I2C_SCL                              | 3V3 level |
|  | 21 | CSI's I2C data signal MIPI_CSI1_I2C_SDA |                                                | 3V3 level |
|  | 22 | Suspended                               | NC                                             |           |
|  | 23 | CSI power enable signal                 | MIPI_CSI1_PWREN                                |           |
|  | 24 | Input power                             | VDD_5V                                         |           |
|  | 25 | GND                                     | GND                                            |           |
|  | 26 | GND                                     | GND                                            |           |

Table 3-1 J7 Pin Definitions

3.2. Camera Interface

The connector's specific model number is CJ-BM4-M11, 4-pin, 18.9\*6\*6mm. It is paired with a QJD-SONY 225 camera: AND: 200W.

Effective pixels: 1920 (H) × 1080 (V).

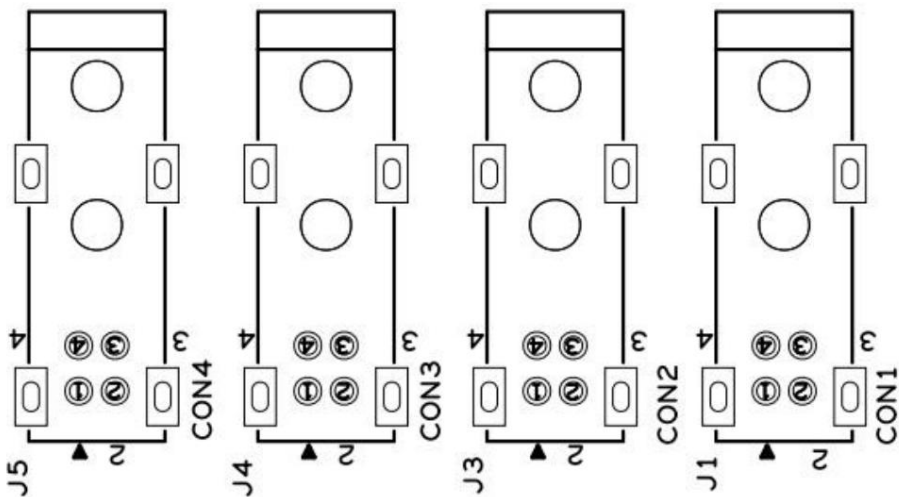


Figure 3-2 Camera Pinout Diagram





3.2.1. Pin Definitions

| Position | logo | Function         | Signal     | illustrate |
|----------|------|------------------|------------|------------|
| J1       | 1    | GND              | GND        |            |
|          | 2    | CSI signal input | VIN-P1     |            |
|          | 3    | Power input      | VDD_CAM_5V |            |
|          | 4    | Suspended        | NC         |            |
|          |      |                  |            |            |
| J3       | 1    | GND              | GND        |            |
|          | 2    | CSI signal input | VIN-P2     |            |
|          | 3    | Power input      | VDD_CAM_5V |            |
|          | 4    | Suspended        | NC         |            |
|          |      |                  |            |            |
| J4       | 1    | GND              | GND        |            |
|          | 2    | CSI signal input | VIN-P3     |            |
|          | 3    | Power input      | VDD_CAM_5V |            |
|          | 4    | Suspended        | NC         |            |
|          |      |                  |            |            |
| J5       | 1    | GND              | GND        |            |
|          | 2    | CSI signal input | VIN-P4     |            |
|          | 3    | Power input      | VDD_CAM_5V |            |
|          | 4    | Suspended        | NC         |            |

Table 3-2 Camera Pin Definitions

3.3. 5V Input Interface

The MY-CAM004M is designed with an external 5V interface J6 to prevent excessive power consumption and the FPC from failing to meet the current requirements.

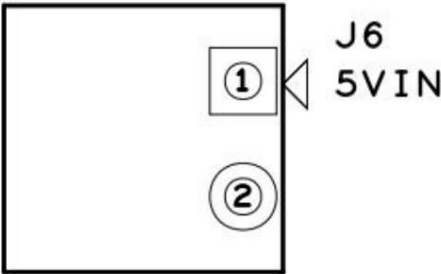


Figure 3 - 3 J6 Pinout Diagram



3.3.1. Pin Definitions

| Position | logo | Function    | Signal | illustrate |
|----------|------|-------------|--------|------------|
| J6       | 1    | Power input | VDD_5V |            |
|          | 2    | GND         | GND    |            |

Table 3 - 3 J6 Pin Definitions

4. Product Parameters

4.1. Specifications

The specifications of MY-CAM004M are as follows:

| Serial Number | Specification             | parameter  |
|---------------|---------------------------|------------|
| 1             | Power supply requirements | 5V         |
| 2             | Power consumption         | 0.8-1.2W   |
| 3             | Operating temperature     | -40-85y    |
| 4             | Pixels                    | 1920x1080P |

Table 4-1 Specifications

4.2. Specifications

The MY-CAM004M module is compatible with the following development boards:

| Serial Number | Product Model | illustrate                           |
|---------------|---------------|--------------------------------------|
| 1             | MYD-YD9X-I    | Requires using the same surface line |
| 2             | MYD-JD9X-I    | Requires using the same surface line |

Table 4-2 List of Supported Products



## 5. Mechanical dimensions

The MY-CAM004MPCB uses a 4-layer board design and measures 65x50x1.6mm.

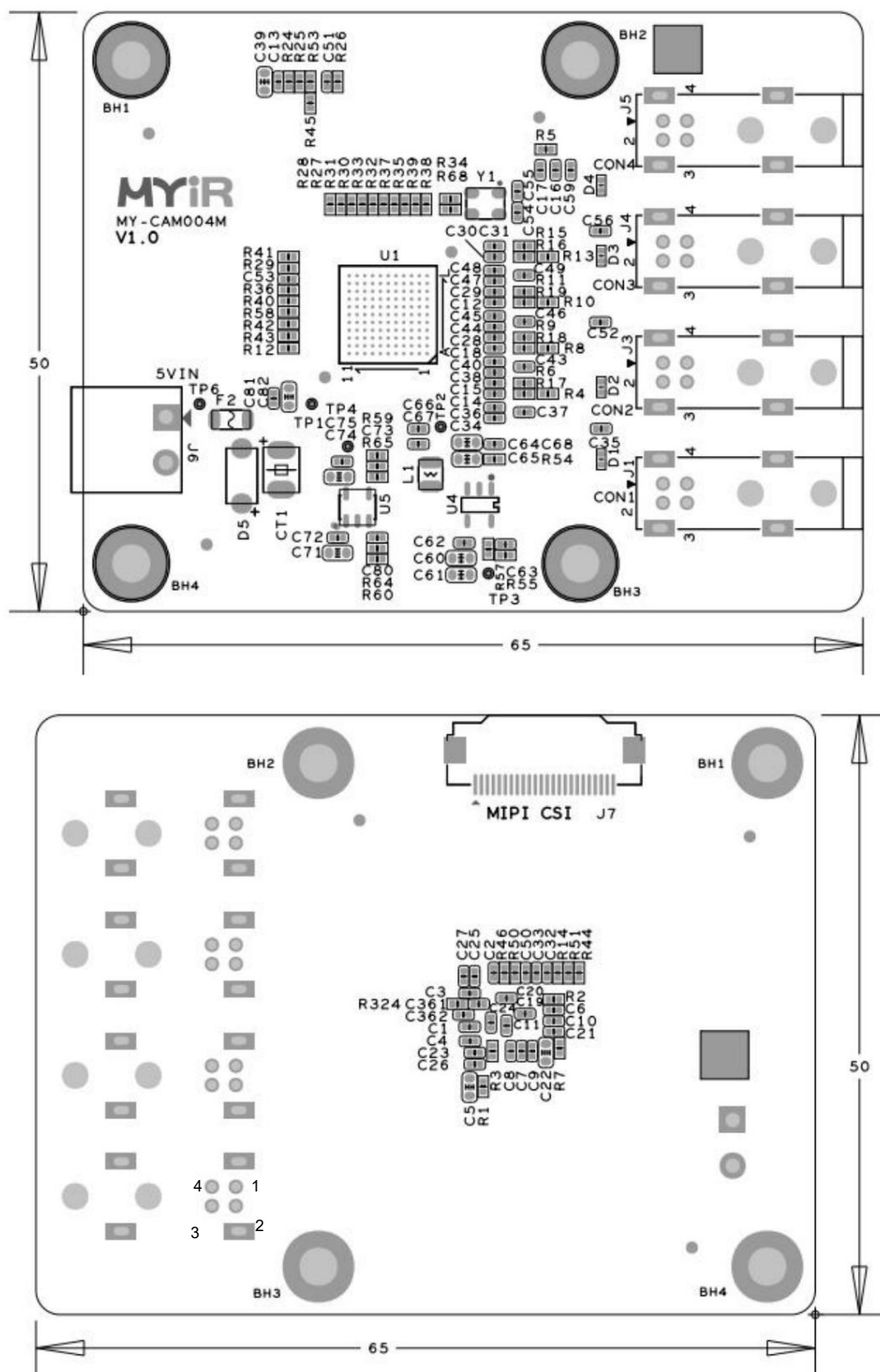


Figure 5-1 Dimensional drawing of MY-CAM004M



## 6. Recommended Companion Camera

一. 产品规格: 1/3 索尼 AHD 200w 高清彩色主板

二. 产品型号: QJD-SONY 225

三. 主要特点:

1. 车载彩色高清摄像模版

2. 3.5-6.5V 宽电压输入

3. 小巧直径

4. 4000V 静电保护

5. 整体性能优良, 画质清晰亮丽

6. 外形尺寸 (直径×高):  $\phi 17\text{mm} \times 8\text{mm}$

四. 规格说明:



|        |                          |                           |
|--------|--------------------------|---------------------------|
| 摄像头器件  | 1/3 索尼 200w CMOS         |                           |
| 系统制式   | PAL                      | NTSC                      |
| 有效像素   | PAL: 1920 (H) × 1080 (V) | NTSC: 1920 (H) × 1080 (V) |
| 最低照度   | 0.01LUX/F1.2             |                           |
| 加马值    | 0.45                     |                           |
| 扫描频率   | PAL: 50Hz                | NTSC: 60Hz                |
| 水平清晰度  | AHD: 200W 像素 1080P       | 模拟: 1600 线                |
| 电子快门速度 | 1/50-1/100000s           | 1/60-1/100000s            |
| 视频输出   | 1.0V p-p AHD             |                           |
| 防水等级   | IP67                     |                           |
| 结构     | 6G 全玻                    |                           |
| 角度     | 182°                     |                           |
| 信噪比    | >52dB                    |                           |
| 操作温度   | -40° C—80° C             |                           |
| 工作湿度   | 低于 90%                   |                           |
| 电源     | DC3.5V-6.5V (±10%)       | 工作电流: 50mA                |



## Appendix 1 Disclaimer

This product manual (hereinafter referred to as the "Manual") will be published as completely and accurately as possible. However, the content is subject to change without notice.

Unless otherwise stated, all company names, personal names, and data used in the examples in this manual are fictitious.

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## Appendix 2 Contact Us

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Telephone: 027-59621648

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Contact Number: 021-62087019 Address: Room

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### Beijing Office

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10, Lipo Plaza, No. 8 Ronghua Middle Road, Daxing District, Beijing





### Appendix 3 Technical Support Instructions

MYIR 's philosophy is "Professional services help developers succeed".

To help customers use our products more quickly and efficiently, MYIR provides comprehensive and thoughtful technical support services through its offices in various locations.

#### Product development materials:

All MYIR development boards come with a data CD, which typically contains the following information:

- Product User Manual
- Product schematic diagram (PDF format)
- Complete example code and BSP package
- Onboard main chip technical manual
- Corresponding development toolchain (GNU tools or third-party tool evaluation board such as MDK)

#### Technical support scope

MYIR provides 6 months of free technical support for the products it sells. The scope of technical support includes:

- The purchased product's hardware and software resources, and hardware warranty.
- Assist customers in correctly using and debugging the sample code provided in the CD-ROM content.

Customer questions regarding product documentation, operation, and the use of embedded software and hardware platforms.

Due to the special nature of embedded development, the following situations are not covered by our free technical support service and will be handled on a case-by-case basis:

- Software and hardware problems encountered by users during self-development, resulting in hardware modifications and damage.

Problems encountered by users in customizing, compiling, and running embedded operating systems

- Programs developed and modified by users themselves on the platform

Problems encountered when modifying the software code on the CD.

For more information about Mier Electronics products, please visit the Mier Electronics website, call or email us. Thank you for your interest in our products.

#### Note!

